

Relativistic quantum antiscars in electrostatically defined graphene quantum dotsYu-Chen Zhuang , Yue Mao , and Qing-Feng Sun ^{*}*International Center for Quantum Materials, School of Physics, [Peking University](#), Beijing 100871, China
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Quantum antiscars characterized by the reduced wavefunction probability density along unstable periodic orbits have been a long-sought subject in the field of quantum chaos. In view of the antiresonance origin, an explicit spatial visualization of quantum antiscars at a single energy level remains still scarce in either theoretical proposals or experimental methods. Here we propose that antiscars can be directly visualized in energy-fixed local density of states (LDOS) spatial maps of electrostatically defined graphene quantum dots (GQDs), as a result of the leakage induced by Klein tunneling. Through systematic numerical simulations, quantum scars and antiscars are demonstrated to alternate between peaks and dips of the LDOS-energy spectrum, collectively following a relativistic equidistant recurrence. A mutual conversion between scars and antiscars can be also achieved by tuning an external magnetic flux. Our work opens horizons for understanding, detecting, and utilizing scarring and antiscarring phenomena in open quantum chaotic systems.

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Introduction. The classical chaos in quantum systems [1,2] gives rise to quantum scars [3] and antiscars [4]. They respectively correspond to wavefunctions exhibiting enhanced and reduced density along unstable periodic orbits [5–7]. Fundamentally, quantum scars and antiscars arise from constructive and destructive quantum interference, providing deep insights into the classical-quantum correspondence of ergodicity [4,8], which extends even to many-body systems [9–12]. The pursuit of quantum scars and antiscars also offers promising applications to regulate electron transport in quantum chaotic devices [13,14]. Up to now, quantum scars have been experimentally detected and spatially visualized in classical wave systems [15–17] and realistic quantum systems [18–20]. Some new members, such as relativistic scars [21,22], chiral scars [23,24], and variational scars [25–28] have also been theoretically proposed to enrich the scarring family. However, experimental visualizations and theoretical characterizations of antiscars are relatively undeveloped.

One key reason for the elusiveness of antiscars is that unlike scars, antiscars are hard to be mapped to specific eigenstates in a closed quantum chaotic system [29]. Generally, bound states formed in closed systems demand wavefunctions to match themselves coherently after surrounding one complete cycle, which seems incompatible with the domination of destructive interference required by antiscars. In view of this, some theories recently proposed to obtain an antiscarring pattern through a collective statistical method, namely, subtracting strongly scarred states from a cumulative density of states [29,30]. Intrinsically, this indirect way is not very surprising, since the cumulative density of states from a summation over a large range of eigenstates should be uniform due to the eigenstate stacking theorem [31]. The appearance

of antiscarring pattern is just a side effect after removing the contribution from quantum scars. This method is a statistical average over an energy window and therefore cannot independently reveal antiscarring behavior at a single energy level. Despite the significant progress made in visualizations of quantum scars, whether an effective platform exists for the direct visualization of antiscars at a single energy level remains an open question.

In this Letter, we theoretically propose that quantum antiscars can be explicitly visualized in local density of states (LDOS) spatial maps at a single energy level in electrostatically defined graphene quantum dots (GQDs). Different from closed quantum dots with strictly discrete bound states, the Klein tunneling makes electrostatically defined GQDs naturally leaky [32], only exhibiting quasibound states with finite lifetimes in a continuous LDOS-energy spectrum [33]. Through systematic theoretical investigations, we show that antiscars can correspond to dips in LDOS-energy spectrum, just beside those resonant peaks demonstrating scars. Due to the linear dispersion of massless Dirac fermions [34,35], the energy interval recurrence of scars and antiscars is approximately equidistant, which is consistent with the relativistic feature. Through varying an external magnetic field, a conversion between scars and antiscars can be also achieved, reflecting a phase regulation via the magnetic flux.

Physical picture of antiscars. We first provide a physical picture to explain why quantum antiscars can emerge in electrostatically defined GQDs. Taking a stadium-shaped billiard as an example [see yellow region in Fig. 1(a)], the classical dynamics in it is nonintegrable and almost all classical periodic orbits are unstable [36]. The ergodicity of classical chaos implies that the measure of unstable periodic orbits in phase space should be zero, which seems to generate a uniform wavefunction distribution [37]. But in quantum chaos, quantum interference effects weakly break this ergodicity [38].

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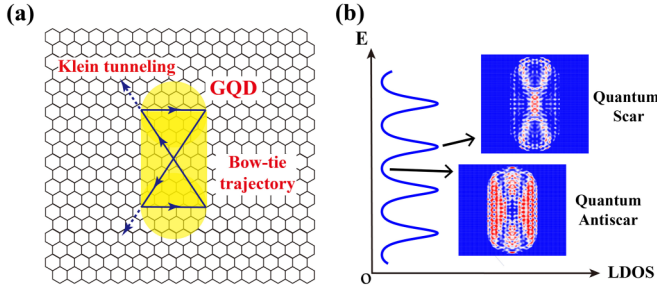


FIG. 1. (a) A schematic diagram of quantum confinement in an electrostatically defined stadium-shaped GQD (yellow color region), which naturally behaves as a leaky quantum dot. The navy-blue solid lines denote a typical bow-tie trajectory where dashed arrow lines denote the Klein tunneling. (b) A schematic diagram for the emergence of quantum scars and antiscars in a continuous LDOS-energy spectrum (dark blue lines). The resonance peaks could exhibit a sign of quantum scars, while the nearby LDOS dips from the antiresonance process could exhibit quantum antiscars.

When the phase accumulation $\Delta\phi$ along a certain unstable periodic orbit arrives at $2\pi m$ ($m = 0, 1, 2, \dots$), the constructive interference may collect the probability density on this orbit, forming a scarring pattern instead of an expected messy pattern [3,21]. Instead, as the phase accumulation $\Delta\phi$ turns to $(2m + 1)\pi$, the destructive interference will suppress the probability density on this orbit, which is expected to lead to an antiscarring pattern [4].

In a closed quantum billiard, electrons are completely confined to form discrete bound states, which requires wavefunctions to satisfy the standing-wave condition after encircling the system. Therefore, scars from the constructive interference can naturally manifest in bound states, while antiscars from the destructive interference will be easily blurred by the domination of other constructive interference processes [29]. The situation becomes quite different in electrostatically defined GQDs. Due to the Klein tunneling [32,39], massless Dirac fermions cannot be completely confined and always have a probability to tunnel out of the GQD along the unstable periodic trajectory (e.g., a bow-tie trajectory [40]); see dashed arrow lines in Fig. 1(a). This indicates that electrostatically defined GQDs behave as special leaky quantum dots rather than completely closed ones. In this case, strictly bound states no longer exist; instead, they are replaced by a series of quasibound states with finite lifetimes [33,41,42]. They will manifest as resonance peaks with finite energy broadenings in a continuous LDOS-energy spectrum [33], and can be resolved by scanning tunneling microscopy/spectroscopy (STM/STS) tip measurements [43–46]. Intrinsically, these peaks of quasibound states also originate from the resonance coherence within the potential cavity [43,47], and have been experimentally verified to manifest quantum scars [20].

The continuous energy spectrum of leaky billiard systems offers an opportunity to observe antiscars [see Fig. 1(b)]. Fundamentally different from completely confined systems with discrete bound states [48,49], the global standing-wave condition is not demanded for electrons in leaky quantum dots. In the continuous energy spectrum, we can investigate the energy positions deviating from those scarred LDOS peaks,

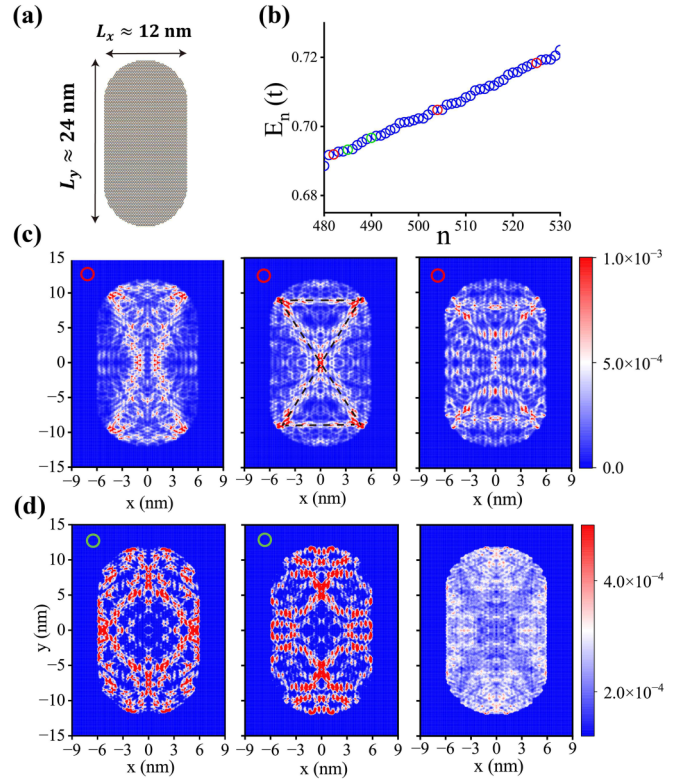


FIG. 2. (a) The schematic diagram for a closed stadium-shaped GQD, with length $L_y \approx 24$ nm and width $L_x \approx 12$ nm. The total number of carbon atoms is 9768. (b) The arrangement of eigenvalues E_n . (c) Three evident quantum scars of bow-tie orbits in LDOS spatial maps, sketched by black dashed lines in middle panel, corresponding to eigenstates $n = 482, 504, 525$ (red circles) in (b), with $E_n \approx 0.6919t, 0.7048t, 0.7184t$. (d) The left two panels denote LDOS spatial maps of eigenstates $n = 485, 490$ (green circles), with $E_n \approx 0.6933t, 0.6967t$. The rightmost panel shows the spatial map of the averaged cumulative LDOS $\bar{\rho}(\mathbf{r})$ summed from $n = 498$ to $n = 510$, but excluding the middle evidently scarred state $n = 504$.

where semiclassical phase accumulation along the concerned unstable classical trajectory will no longer match the constructive interference condition $\Delta\phi = 2\pi m$. Generally, there is a LDOS dip between two adjacent LDOS peaks, exhibiting a pronounced suppression with a minimum LDOS value [see Fig. 1(b)]. The strong suppression indicates an antiresonance process experienced by electrons, which may satisfy the domination of destructive interference along the concerned unstable periodic orbit [$\Delta\phi = (2m + 1)\pi$]. Therefore, LDOS dips between scarred LDOS peaks may exhibit antiscars.

Identification of antiscars. To validate the proposed physical picture, we now construct a tight-binding model to conduct numerical simulations. The numerical simulation details can be found in Sec. SI in the Supplemental Material [50] (including Refs. [51–53]). As a comparison, we will first demonstrate the obscurity of antiscars in a closed GQD, then present their direct visualizations in electrostatically defined GQDs. The closed GQD with width $L_x \approx 12$ nm and length $L_y \approx 24$ nm [see Fig. 2(a)] is directly truncated from one graphene sheet [21], and has been extensively investigated before [54,55].

The electrostatically defined GQD with a similar size is achieved by adding potential $V(\mathbf{r})$ on the graphene sheet [56]. In experiments, such electrostatically defined GQDs can be implemented multiple ways, like using the effect of the substrate [41,44–46] and STM-tip bias-voltage pulsing [42,57].

The eigenvalues E_n of the closed stadium-shaped GQD are plotted in Fig. 2(b). The index n is counted from the first positive eigenstate here. The energy unit is chosen as the nearest-neighbor carbon-carbon hopping strength $t = 2.8$ eV. Note that we focus on a relatively high-energy region ($E \approx 0.7t$), where the electron wavelength λ [58] is far smaller than the size of the system ($\lambda < 2$ nm). In this case, the semiclassical dynamics of electrons in GQDs will be more evident (i.e., classical limit). In Fig. 2(c), the LDOS spatial maps of three scarred eigenstates Ψ_n with $n = 482, 504, 525$ are shown [red color states in Fig. 2(b)], each exhibiting an evident scarring pattern of the bow-tie trajectory (sketched by black dashed lines). From the semiclassical perspective [54,59,60], they should correspond to the successive completion of $2m\pi$ phase accumulation along the unstable bow-tie trajectory. Due to the linear dispersion of massless Dirac fermions, these relativistic quantum scars share an approximately equal energy interval about $0.0133t$ [21], which is roughly consistent with the theoretical estimation: $\Delta E \approx 2\pi \frac{\hbar v_F}{L} \approx 0.0157t$. Here the orbit length is estimated about $L \approx 66$ nm with the naked eye (see theoretical estimation details in Sec. SI in the Supplemental Material [50]).

However, none of the eigenstates exhibit clear signatures of quantum antiscars in their LDOS spatial maps. As an example, the left two panels of Fig. 2(d) show two eigenstates ($n = 485, 490$) between the scarred states ($n = 482, 504$) [green color states in Fig. 2(b)]. The LDOS spatial distributions of them do not exhibit expected antiscarring patterns of the bow-tie orbit. Similar phenomenon can be also verified in the other nineteen eigenstates between $n = 482, 504$, which mainly exhibit the characteristics of ergodic states with messy patterns [61], or scarringlike states possibly from coherence of other single or multiple classical orbits [8], (see Sec. SII and Fig. S1 in the Supplemental Material [50]). In the rightmost figure in Fig. 2(d), we further present an averaged cumulative LDOS $\bar{\rho}(\mathbf{r})$ for a summation of eigenstates from $n = 498$ to $n = 510$, but excluding the strongly scarred eigenstate $n = 504$ in the middle (see definitions in Sec. SI in the Supplemental Material [50]). It demonstrates an approximate antiscarring pattern of the bow-tie trajectory. As a result of eigenstate stacking theorem [29,30], the summation of eigenstates over a relatively large energy window should be uniform [31]. Once subtracting those obvious scarred states, the remaining cumulative LDOS will finally exhibit antiscarring phenomena. This method implies quantum scars and antiscars are inherently complementary parts under the interplay of quantum coherence and classical chaos.

As shown in Fig. 2, quantum antiscars are obscure at the single eigenstate level in closed GQDs due to the constraints of standing-wave conditions for bound states. Beyond this limitation, the domination of resonance and antiresonance processes can both appear in leaky GQDs shown in Fig. 1, offering possibility to identify antiscars.

Based on the graphene tight-binding model, the LDOS space-energy map along the long axis (i.e., y axis) of the elec-

trostatically defined stadium-shaped GQDs is demonstrated in Fig. 3(a). Due to the confinement of the potential field, numerous bright antinodes emerge within the GQD, reflecting the existence of quasibound states [44,46]. Specifically, the LDOS-energy spectrum at the center of the GQD (black dashed line) is plotted in Fig. 3(b), where the spectrum is continuous and the LDOS is finite at any given energy. Still focusing on a high-energy range relative to the GQD potential platform ($E \approx -0.6t$), we also capture scars of bow-tie trajectories in LDOS spatial maps at energy positions of some resonance peaks [see Figs. 3(c) and 3(e)]. Compared to those in Fig. 2(c), scarring patterns under the electric confinement are closer to the lemniscate ∞ form, which is consistent with observations in the related experiment [20].

Notably, at the energy positions of neighboring LDOS dips [green dashed lines in Fig. 3(b)], the characteristics of LDOS spatial distributions are completely reversed. The original high-density regions are replaced by low-density regions, demonstrating typical quantum antiscars of bow-tie trajectories. A representative signature to distinguish quantum scars and antiscars is the high versus low density at the GQD center, which exactly accounts for the LDOS peaks and dips in the LDOS-energy spectrum in Fig. 3(b). In fact, the phenomenon of successive quantum scars and antiscars can persist across a wider range of energy (see Sec. SIII and Fig. S2 in the Supplemental Material [50]), reflecting their robustness. The proximity of quantum scars and antiscars suggests that they should correspond to two adjacent processes for resonance $\Delta\phi = 2m\pi$ and antiresonance $\Delta\phi = (2m+1)\pi$. To reveal this, we investigate the energy arrangement of a series of quantum scars and antiscars and find they collectively exhibit a quasiequidistant spacing with the energy interval about $0.0094t$ (see Sec. SIII and the inset in Fig. S2(a) of the Supplemental Material [50]). Similar to relativistic quantum scars, we can refer to the shown quantum antiscars as “relativistic quantum antiscars.” According to the pattern in Fig. 3(c), we estimate the length of the bow-tie orbit about $L \approx 57$ nm with the naked eye. It is slightly shorter than the orbit length in Fig. 2(c) due to a slight change of the scarring/antiscarring pattern. Based on the semiclassical method, this interval is still consistent with the theoretical estimation $\Delta E \approx \pi \frac{\hbar v_F}{L} \approx 0.0091t$, which corresponds to a phase difference of π rather than 2π . Moreover, besides the bow-tie trajectory, we also find the similar signature of quantum scars and antiscars with a diamond-shaped trajectory [21] at a lower energy (see Sec. SIV and Fig. S3 in the Supplemental Material [50]).

These above results reveal that leakage plays an important role in the manifestation of antiscars by lifting the restriction imposed by the global standing-wave condition in closed systems. In principle, other types of leaky quantum dots [62–65], may also demonstrate similar antiscars. To clarify this, we perform numerical calculations on a stadium-shaped quantum dot defined on a square lattice with the introduction of leakage. We systematically investigate the dependence of the antiscarring on the leakage strength, and also observe explicit antiscars at appropriate leakage strengths. These results reflect the universality of our findings (see more details in Sec. SV and Figs. S4–S17 in the Supplemental Material [50]).

Magnetic-flux-induced conversion between scars and antiscars. As scars and antiscars correspond to two different

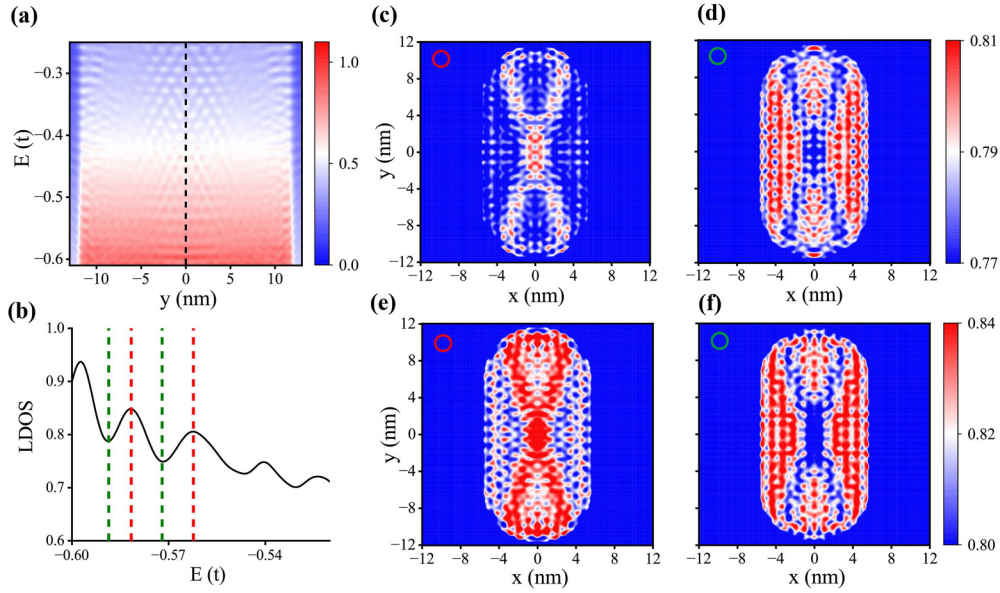


FIG. 3. (a) The LDOS space-energy map along the long axis (y axis) of the electrostatically defined GQD. (b) The LDOS-energy spectrum at the center of the GQD, which corresponds to the black dashed line in (a). (c), (e) The LDOS spatial maps for resonance peaks [red dashed lines in (b)], with $E \approx -0.5623t$, $-0.5816t$. (d), (f) The LDOS spatial maps for LDOS dips [green dashed lines in (b)], with $E \approx -0.5720t$, $-0.5886t$. Note that (c) and (d) are the same as figures shown in Fig. 1(b).

phase coherence processes, a transformation between them is possibly achieved through a phase regulation. When an external magnetic field is applied, an additional magnetic flux $\Phi = B \cdot S_A$ will be introduced [66,67]. Here B is the magnetic field strength and S_A is the area enclosed by the orbit over one complete cycle. Along one fixed periodic orbit, the semiclassical phase accumulation is modulated as [67] $\Delta\phi(B) = \Delta\phi_0 + 2\pi \frac{\Phi}{\Phi_0}$ (also see derivation details in Sec. SI in the Supplemental Material [50]). Here $\Delta\phi_0$ is the initial phase accumulation without B and Φ_0 is the flux quantum. It is likely to achieve a π -phase shift by adjusting B , causing a conversion between scars and antiscars. To show this, we respectively study the magnetic responses of quantum scars/antiscars of the bow-tie trajectory and the diamond-shaped trajectory. Because the area enclosed by the bow-tie trajectory cancels to $S_A = 0$, no essential alterations happen for this kind of antiscars (or scars) even if the magnetic field increases to $B = 10$ T, 20 T in Fig. 4(a). Whereas for the diamond-shaped trajectory, its enclosed area is finite: $S_A \approx 132 \text{ nm}^2$. Once the magnetic field B goes up to about 16 T, an additional π phase is added to the $\Delta\phi(B)$. Note that for a larger GQD, the required B to tune scars to antiscars should be lower. In Fig. 4(b), two LDOS spatial distributions showing the quantum scars and antiscars of the diamond-shaped trajectory (black dashed lines) under $B = 0$ are shown. As the magnetic field climbs to $B = 15$ T, the original scarring pattern with high LDOS density along the diamond-shaped trajectory is replaced by the anticarring pattern with the low LDOS density [following yellow arrows in Fig. 4(c)]. Further increasing the magnetic field to $B = 30$ T, the antiscar returns to the quantum scar. A similar process also holds for the antiscar in the right panel of Fig. 4(b) (see Sec. SVI and Fig. S18 in the Supplemental Material [50]). This magnetic-flux-induced conversion between scars and antiscars again demonstrates that scars and antiscars

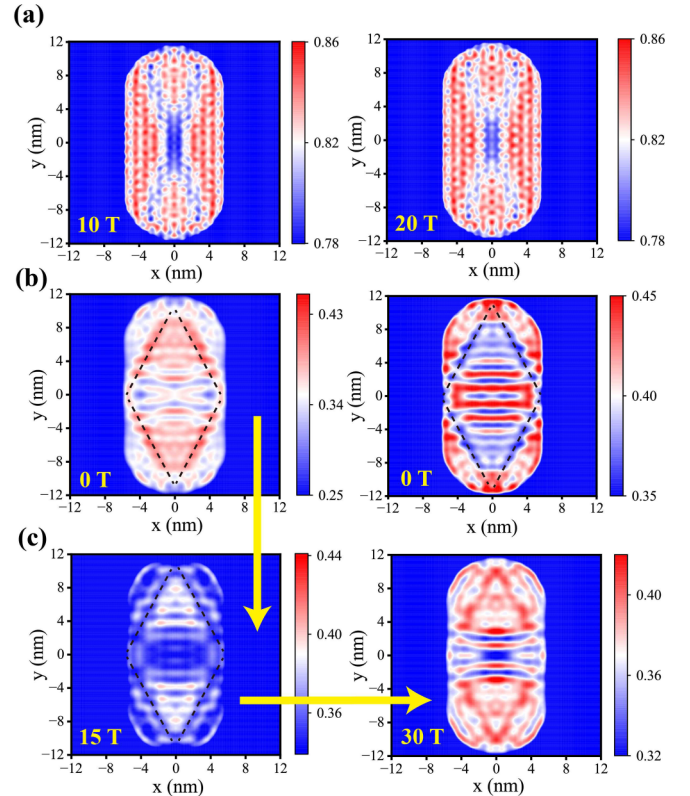


FIG. 4. (a) The quantum antiscars for Fig. 3(f) under $B = 10$ T, 20 T. (b) The LDOS spatial maps for quantum scars (left panel) and antiscars (right panel) of the diamond-shaped trajectory, which are schematically denoted by black dashed lines. Their energies are $E \approx -0.2505t$, $-0.2607t$, respectively. (c) Following the yellow arrows, the quantum scar for $E \approx -0.2505t$ at $B = 0$ T transforms to the antiscar at $B = 15$ T, and returns to the quantum scar at $B = 30$ T.

arise from an opposite phase coherence process of classical trajectories. It also offers a routine to switch electron movements in quantum chaotic systems.

Conclusions. In summary, we propose that the electrostatically defined GQDs can serve as an excellent platform to directly visualize quantum antiscars. Unlike the previous method of simple statistical summation of discrete eigenstates over a certain energy interval, our work catches antiscars at a single energy level, intrinsically reflecting the energy selection among different resonance processes. We find scars and antiscars are respectively related to LDOS peaks and dips in the continuous LDOS-energy spectrum, and overall follow an equally spaced energy distribution. This indicates that the dips in the LDOS spectrum may contain physical information as rich as that of peaks, enlightening subsequent studies of quasibound states in various open quantum dots. By applying an external magnetic field, a mutual conversion between quantum scars and antiscars can be achieved. Considering the mature experimental fabrication of electrostatically defined GQDs in various configurations [46,57,68,69], the availability of high-precision STM/STS measurements [39,42,44,66,70], and especially the first explicit experimental visualization of

quantum scars in such systems [20], our scheme holds great promise for further experimental verification. Moreover, the conversion between scars and antiscars is promising in novel applications of graphene-based electron optics [71], such as magnetic-tunable directional guiding and switching of the electron waves in chaotic nanodevices.

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Data availability. There are no publicly available research data or software supporting this manuscript. Requests for further information or data should be sent to the authors.

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